4.5. Damping of cosmological perturbations

4.5.1 Damping during Recombination

As Universe expands density & Temp of photon - baryon fluid drop. Protons capture e^- and form neutral H. $\lambda_{mfp} \approx \frac{1}{\sigma_T n_e}$ where n_e is number density of free e^- and σ_T is Thomson Scattering Cross section. $\lambda_{mfp} \xrightarrow{n_e \to 0} \infty$ and Universe becomes transparent to radiation.

Recombination happens around $z \sim 1100$ n_e plummets over redshift change $\Delta z \sim 80$ During this time photons will be able to random walk out of overdensities as they scatter off e^-. They will drag matter from over dense to under dense regions and thus damp pert. on scales smaller than characteristic random walk scale.

Silk damping scale.

Consider a photon that suffers $N = c\Delta t / \lambda_{mfp}(t)$

Collisions during recombination - Between 2 collisions it travels a comoving distance $\frac{\lambda_{mfp}}{a(t)}$

So mean square comoving displacement

$$(\Delta x)^2 = N\left(\frac{\lambda_{mfp}}{a}\right)^2 = c\,\lambda_{mfp}(t)\frac{\Delta t}{a^2(t)}$$

So total mean square comoving distance it travels

$$x^2 = \int_0^{t_*} \frac{c\,dt}{\sigma_T n_e(t) a^2(t)} = \int_0^{\eta_*} \frac{c\,d\eta}{\sigma_T n_e(\eta) a(\eta)}$$

$$= \frac{c\,\eta_*}{5\,\sigma_T n_e[\eta_*]\, a[\eta_*]}$$

using: recombination is during matter dom. epoch

$$a(\eta) = a(\eta_*)\left(\frac{\eta}{\eta_*}\right)^2$$

$$n_e(\eta) = n_e(\eta_*)\left(\frac{a(\eta)}{a(\eta_*)}\right)^{-3}$$

$$= n_e(\eta_*)\left(\frac{\eta}{\eta_*}\right)^{-6}$$

Comoving Silk damping scale is:

$$\lambda_S^c = x = \sqrt{\frac{c\,\eta_*}{5\,a(\eta_*)\sigma_T n_e(\eta_*)}}$$

η_* is conformal time @ decoupling.

$\longrightarrow \lambda_S^C \simeq 9.2$ Mpc.

Fluctuations below this scale will be washed out.

4-5-2 - Massive Neutrinos during any epoch

No fluid description as no interaction -
On very large scales they will tend to cluster just like matter & radiation but on small scales they will free-stream (move relativistically) from one region of space to another - This leads to an overall damping effect on free-streaming scale. which depends on mass of neutrino.

$$\lambda_{FS} \simeq 28 \left(\frac{m_\nu}{30eV}\right)^{-1} \text{Mpc}.$$

5 - Building models of Large scale structure (LSS).

Need initial conditions for perturbations.
Goal is to identify how we should characterize LSS today.

$\lambda(t) = \lambda_0 \frac{a(t)}{a_0}$, non-rel. matter

decreases as $\rho_M(t) = \rho_0 \left(\frac{a_0}{a(t)}\right)^3$

$$M(\lambda_0) = \frac{4\pi}{3} \rho_M(t) \left(\frac{\lambda(t)}{2}\right)^3$$

$$= \frac{4\pi}{3} \rho_M^0 \left(\frac{\lambda_0}{2}\right)^3$$

$$= 3 \times 10^{11} M_\odot \left(\frac{\lambda_0}{1\,\mathrm{Mpc}}\right)^3$$

$\lambda_0 \sim 1$ Mpc $\longrightarrow$ mass of galaxy

Point at which $\lambda(t) = R_H(z)$

$$\longrightarrow z_{enter} \simeq 4.55\, 10^5 \left(\frac{\lambda_0}{1\,\mathrm{Mpc}}\right)^{-1}$$

Inflation: • quantum fluctuations are stretched to scales larger than $R_H(z)$

- exponential expansion is driven by a 'cosmological constant' like eq. of state $p < 0$.

Deeply appealing feature:

Random field with Scale invariant gravitational potential.

Theory won't tell you where galaxies are on the sky: we'll talk about how much more likely a structure of a given size will be relative to others.

δ (or grav. pot) is random field for which we will calculate statistical properties.

Average: $\langle \delta(\vec{x}) \rangle = 0$

Variance: $\langle \delta(\vec{x}) \delta(\vec{x}+\vec{r}) \rangle \equiv \xi(\vec{r})$

ξ is auto-correlation function

$$dP = \rho_0^2 \left[1 + \xi(\vec{r})\right] dV_1 dV_2$$

↑ Prob. of finding 2 galaxies in volumes

dV_1 & dV_2 separated by distance r.

NB: $\langle\ \rangle$ indicate volume average.

⟶ deep pb be we want ensemble average

Best we can do is look at widely separated parts of space ⟶ assume ergodicity to say average over space = ensemble average.

Since δ is real, that means that

$$\delta(\vec{r}) = \frac{V}{(2\pi)^3}\int |\delta_k|^2 e^{-ik\vec{r}} d^3k$$

So we will mostly talk about the

Power Spectrum $\quad P(k) \equiv \langle |\delta_k|^2 \rangle$

This 2-point statistics completely defines our random field provided it is Gaussian which is another appealing feature of inflation. Otherwise we would need higher order statistics.

Useful to consider dimensionless version of $P(k)$ called the mass variance:

$$\Delta^2(k) \equiv \frac{V}{(2\pi)^3} 4\pi k^3 P(k)$$

. Scale invariance of ϕ

Define avg grav. pot in ball of radius R and volume V_R:

$$\phi(R) = \frac{1}{V_R} \int_{V_R} d^3x \, \phi(\vec{x})$$

variance: $\sigma_R^2(\phi) = \langle \phi^2(R) \rangle$

A scale invariant spectrum corresponds to a variance of ϕ indep. of R.

$$\sigma_R^2(\phi) = \underline{\underline{cst}}$$

Newton - Poisson to relate $\sigma_R^2(\phi)$ to $\delta(t, \vec{k})$

$$\sigma_R^2(\phi) = \frac{k^3}{4\pi} \langle |\phi(t, \vec{k})|^2 \rangle \propto \frac{k^3}{k^4} \langle |\delta(t, \vec{k})|^2 \rangle$$

with $k = \frac{2\pi}{\lambda}$

So scale invariant means $P(k) \propto k$

This often corresponds to picking amplitude

of $|\delta(\vec{k},R)| \propto R$ (although <0 / t^{ν} in that case but since we care about $P(k)$ and not $\delta(\vec{k},t)$ today does not matter)

Now that we have initial conditions look at LSS for $\neq^{t}$ models of Universe

BDM, CDM, HDM.